

1. Explain the similarities between an array and a stack
They both store multiple items
Both allow access to elements, but in a stack, access is restricted to the top element.
Both organize data in a structured order
2. Describe how a programmer might use a stack when writing compiler software.
A programmer can use a stack to store data so the last item added is the first one removed (LIFO)
The programmer can push items to the stack and pop items and using isEmpty to check
3. What output is generated by the following statements if s is a Stack object?
8 13
8 12
4. Consider the “hot plate” problem: In a busy restaurant, fresh salad plates are brought out and added to the existing pile of plates. This means that, even though there may be plates that were washed long ago, the customer is frequently stuck with a warm plate for salad. Explain how this is analogous to a stack.
 - Plates are added to the top of the pile
 - Customers take the top plate
 - The most recent plate added is the first removed
 - LIFO just like Stack
5. What output is generated by the following statements if q is a Queue object?
5 8
12 5
6. Explain the difference between a FIFO and a LIFO data structure.
First in First out (FIFO) Queue: The first item added is the first removed
Last in First out (LIFO) Stack: The last item added is the first removed
7. List two real-world situations that could be represented as a queue structure other than those listed in the chapter.
 - Cars waiting in a drive-through: The first car in is the first car to get served
 - Customers waiting to checkout: First customer in is the first to pay
8. a) A stack data structure has a front and a rear.
False, a stack only has a top

b) A stack can be emptied.
True

c) In a stack, top refers to the first item pushed onto a stack.
Top refers to the most recent pushed item, not the first one

d) The isEmpty operation returns an int value.
False, it returns a boolean value

e) A queue can hold more than one data item.
True

f) In a queue, all removals are made at the rear
Removals are made at the front

g) The enqueue operation adds an item to the front of the queue
Enqueue adds an item to the rear of the queue

h) The first item in a linked list is called the head
True

i) A node refers to an item in a stack.
False, a node is part of LinkedList not a stack

j) The number of items in a stack or queue can be determined with the length operation.
False, the operation used is size

k) In a linked list, the tail points to null.
True